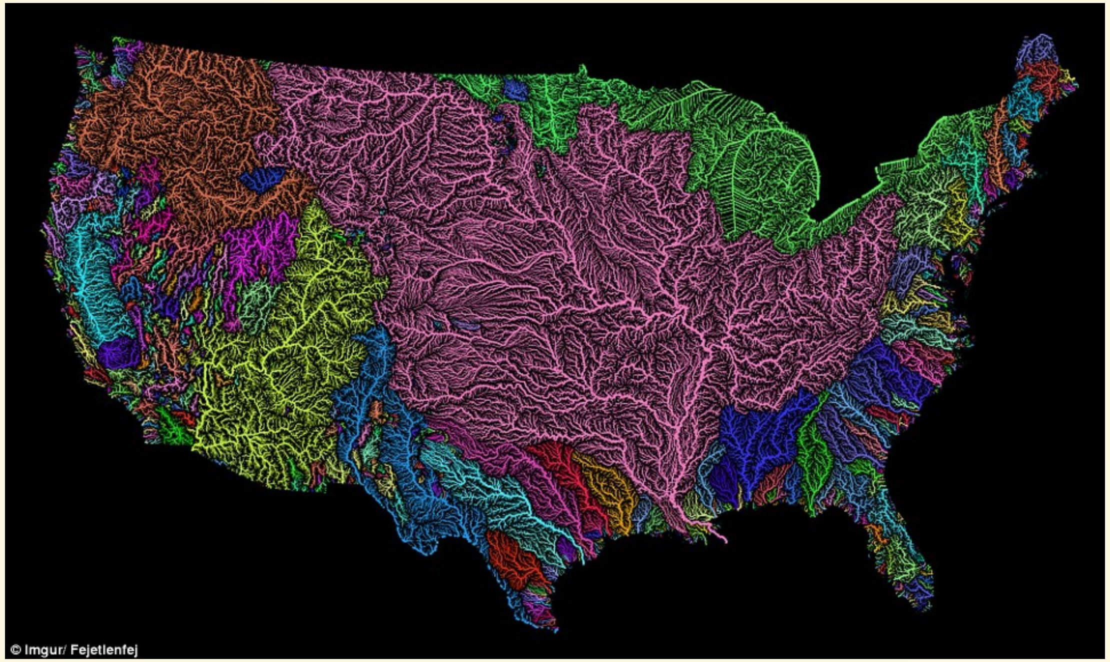
ECO 331

Economic History

GEOGRAPHY

Jonathan Conning
Spring 2026



Readings

- OG 10 "The Shadow of Geography"
- KR 2 "Did Some Societies Win the Geography Lottery" (19-36)
- National Geographic (2015) *Guns, Germs, and Steel*. Episode 1 of 3, especially 29:41-end. (video link.)

Fundamental versus Proximate Causes of Growth

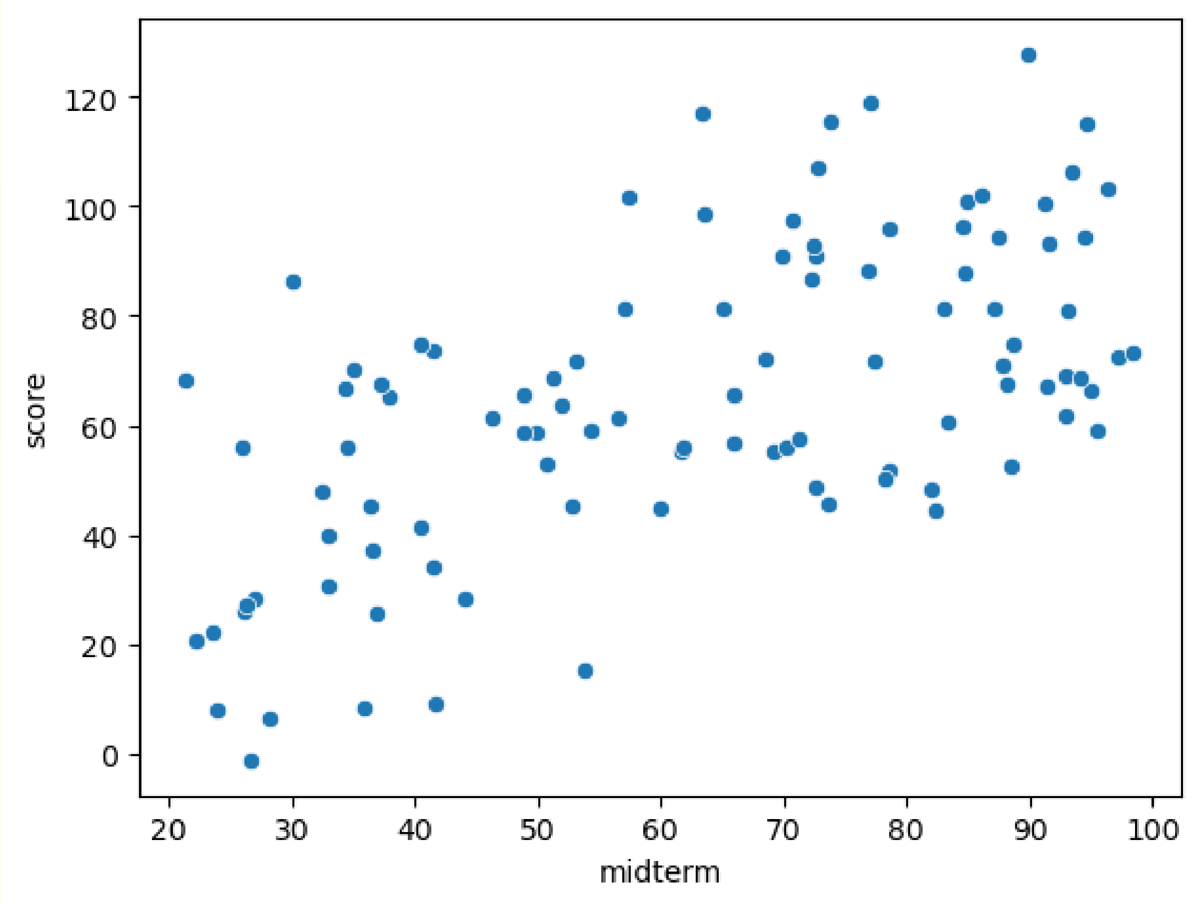
- Proximate Causes
    - Capital and Human Capital Accumulation
    - Trade and Specialization
    - Technological Innovation
- But what causes these things to happen?
- Candidates: Geography, Institutions, Culture, Demography.

Geography Questions

- What geography features constrain/enable economic growth?
  - What are the causal mechanisms?
  - What is Jared Diamond's main thesis?
- Is geography fate? Can transport infrastructure reverse effects?
- Do places that require more cooperation to overcome environmental hardships have an advantage or disadvantage?
- Does better geography raise the level and/or the growth rate?

A simple primer on linear regression

Suppose we have data on students scores on a midterm and a final.



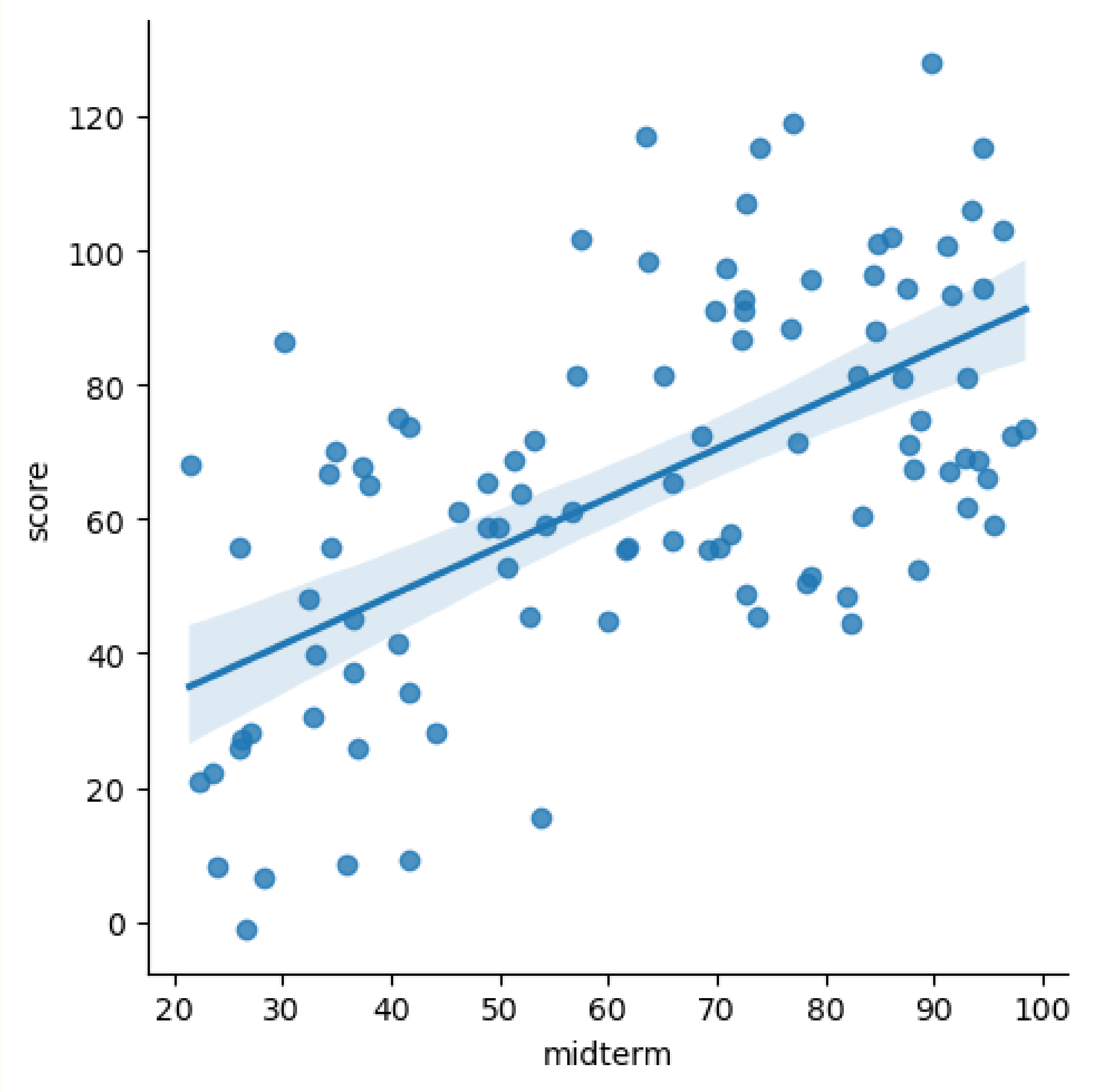
We hypothesize

Linear Regression to fit a hypothesized linear relation of the form

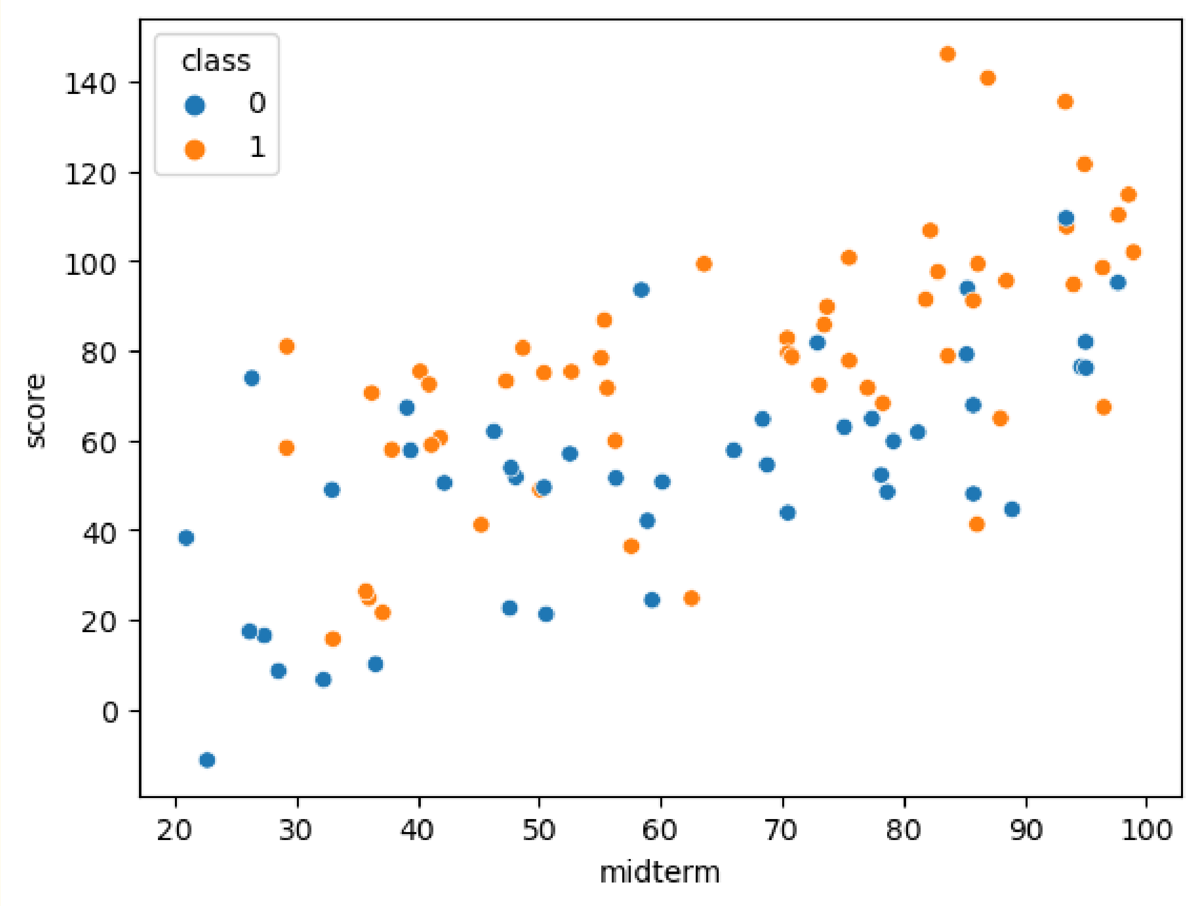
$$y_i = a + b_1 \cdot x_i + e_i$$

where:

y_i = final exam score
x_i = midterm exam score
e_i = i.i.d. error



Students w/ high attendance (orange) have on avg. higher scores:

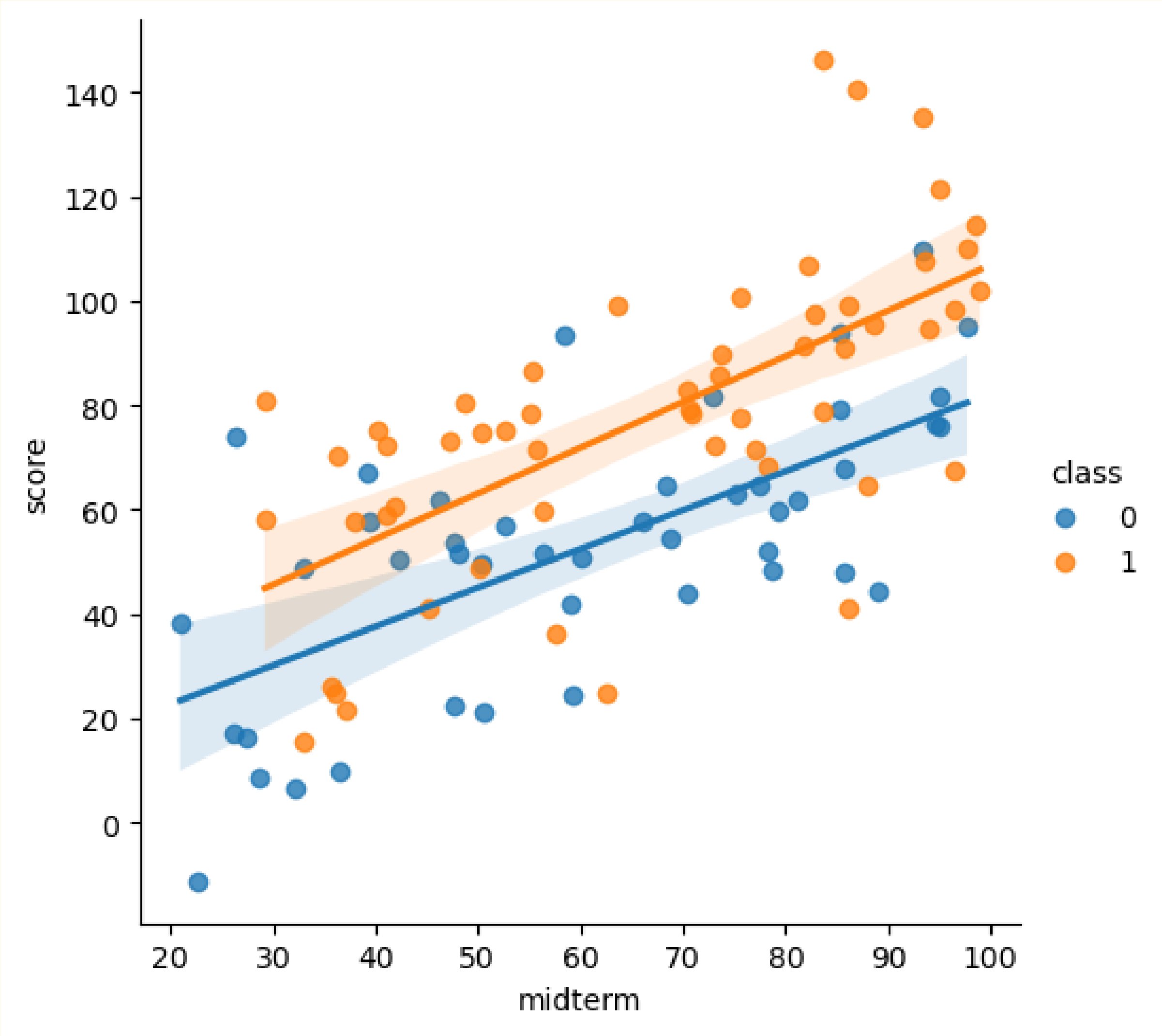


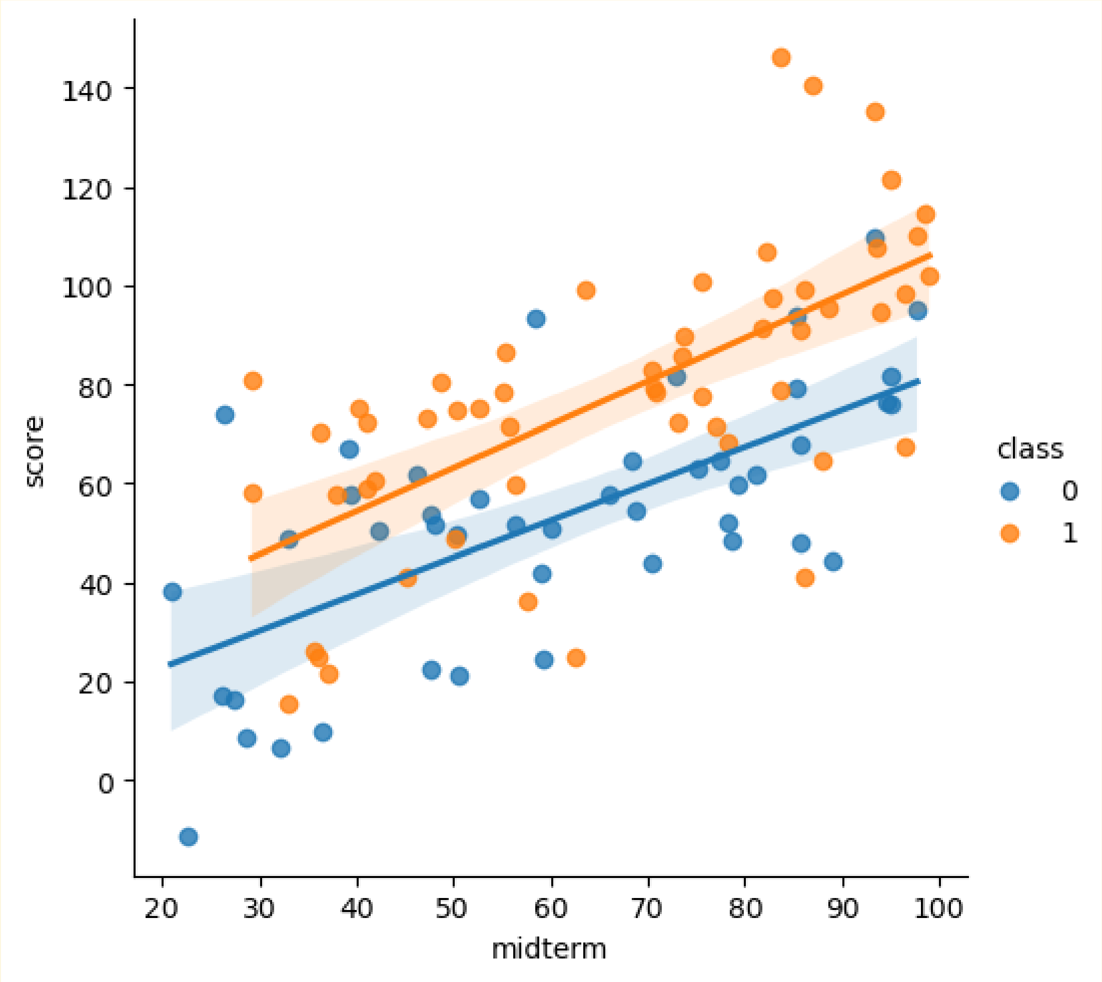
Revised model

- Controls for a previously omitted variable

$$y_i = a + b_1 \cdot x_i + b_2 \cdot D_i + e_i$$

y_i = final exam score
x_i = midterm exam score
D_i = 1 if attended regularly
e_i = error term

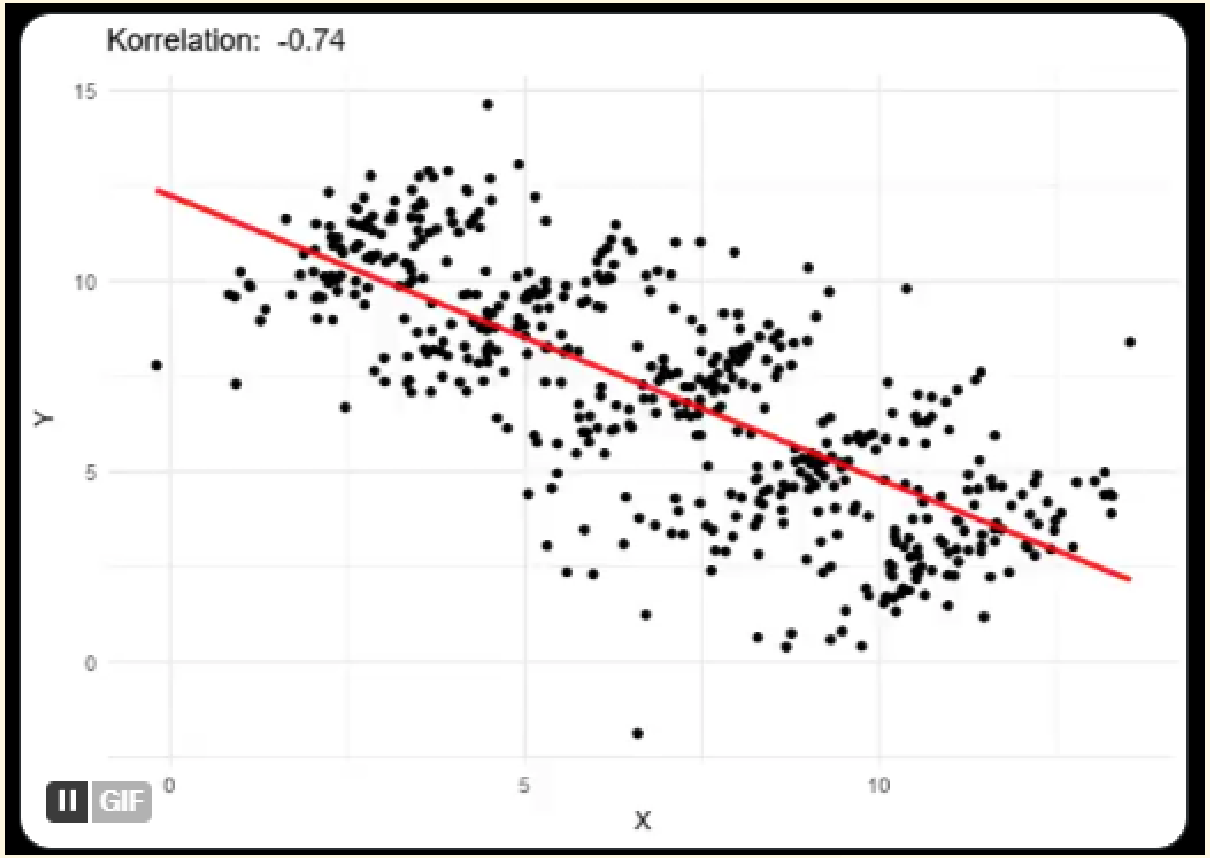


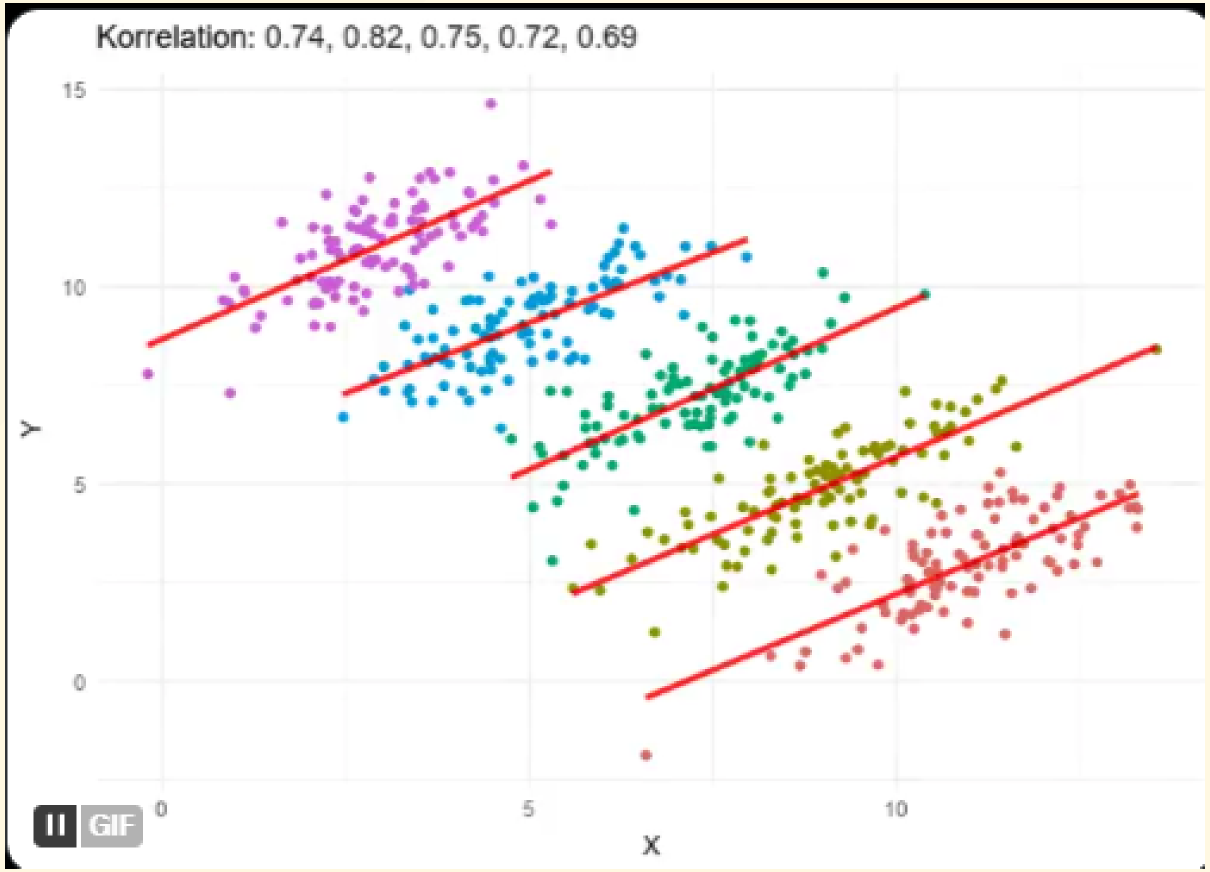


	coeff	se	t	p>t
midterm	0.813	0.094	8.686	0.000
class	19.894	4.212	4.723	0.000
_cons	3.637	6.477	0.562	0.576

If `class` dummy is omitted; estimated b_1 coefficient biased upward.

	coeff	se	t	p>t
midterm	0.873	0.102	8.532	0.000
_cons	10.734	6.951	1.544	0.126


Korrelation: -0.74
15
10
5
0
Y
0
5
10
X
GIF

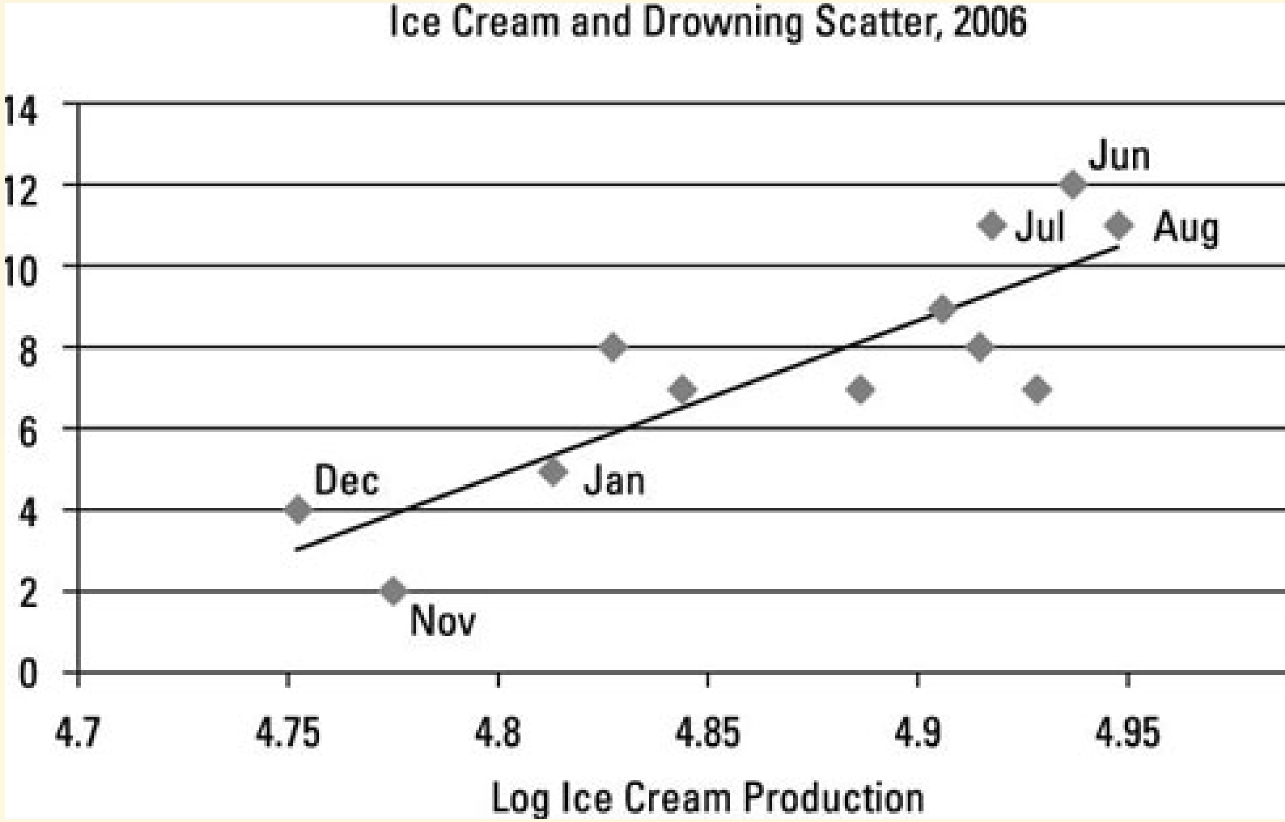

Korrelation: 0.74, 0.82, 0.75, 0.72, 0.69
15
10
5
0
Y
0
5
10
X
GIF

Econometric Challenges

- Identification challenges: are estimated effects causal and unbiased.
- Spurious correlation or Causation?
  - Are the regressors exogenous or Endogenous?
- Biased estimates might be due to:
  - omitted variables
  - selection or endogeneity bias
    e.g. is x_i variable exogenous (like the weather) or determined by same unobserved factors that affect y_i

Does ice-cream explain pool drownings?

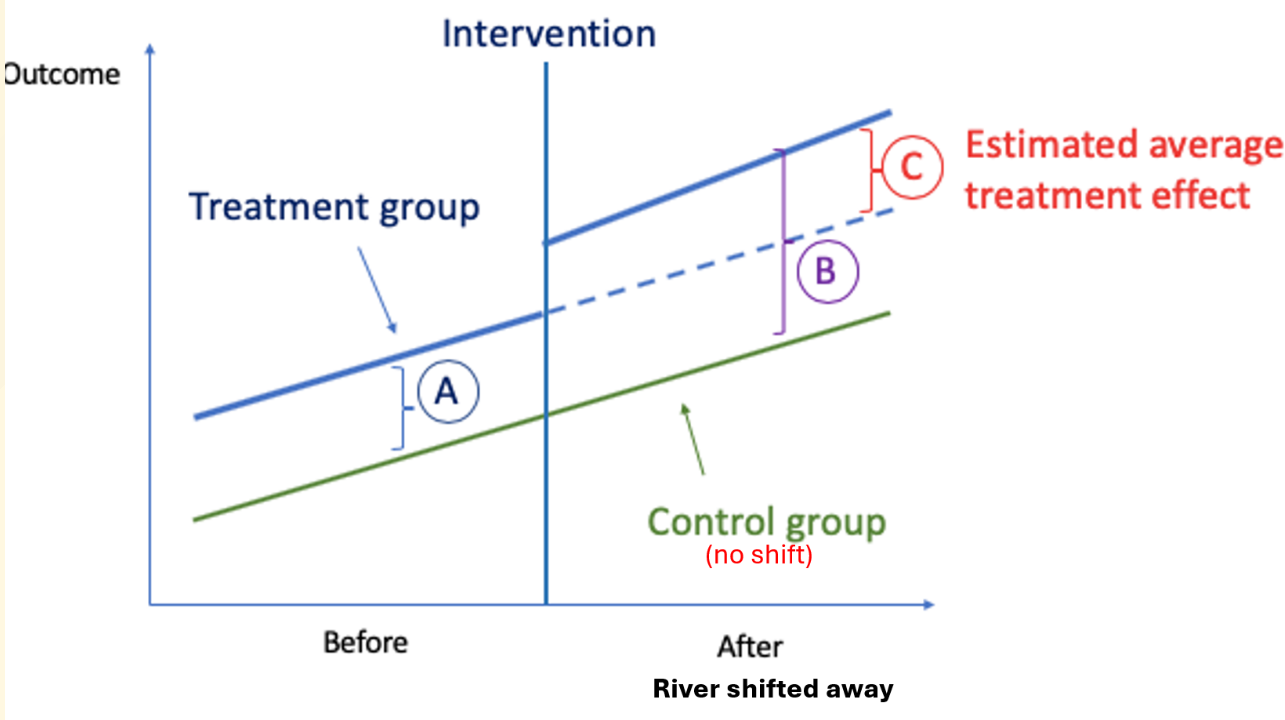
- ice-cream consumption jointly determined with pool drownings
- both are affected by temperature
    - when weather is hot
        - more pool use and drownings
        - more ice-cream consumption

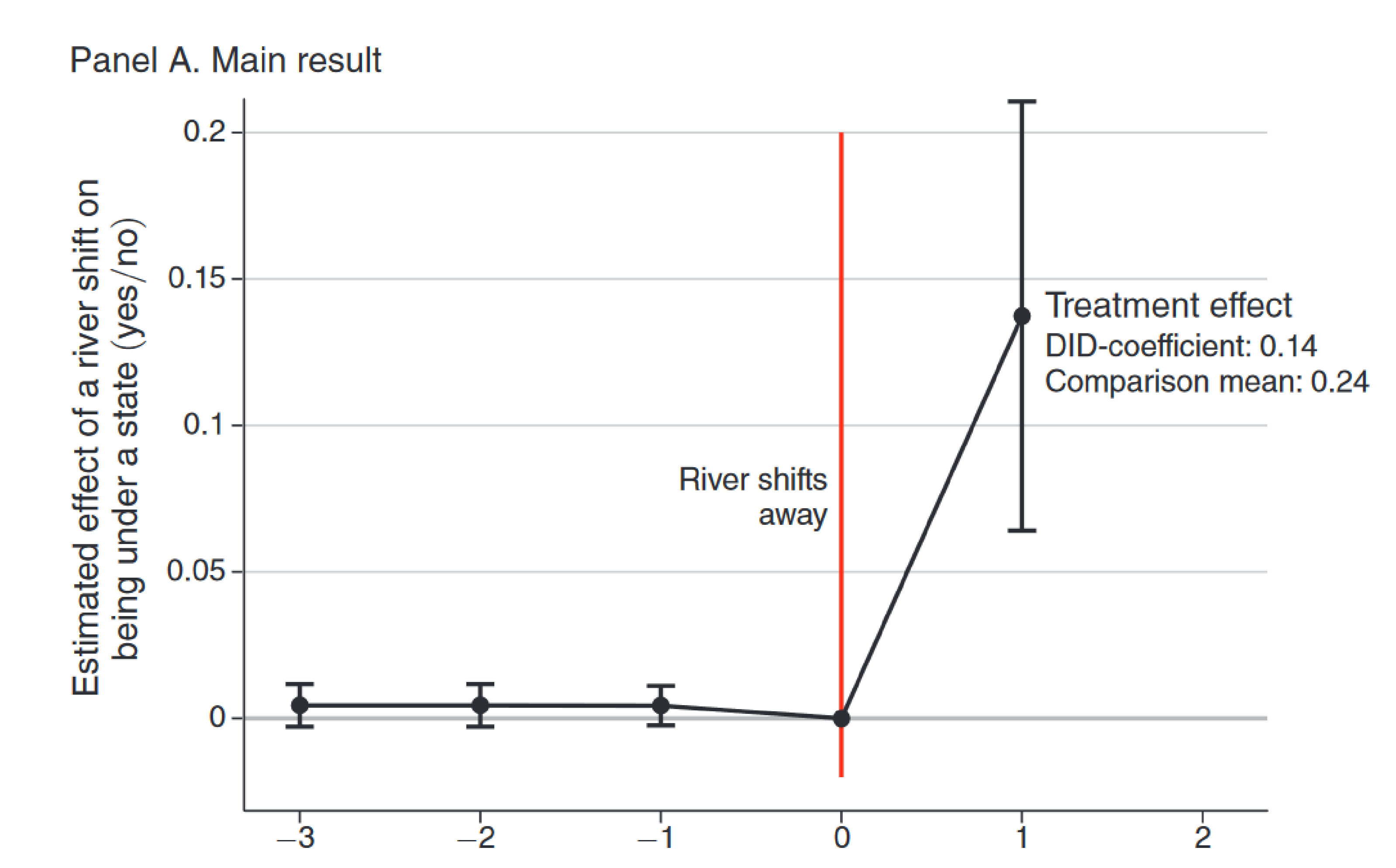


Empirical Methodology: Difference-in-Differences

Recall Allen et al (2023) “The Economic Origins of Government.”

- 1374 grid cells covering Tigris-Euphrates area
- 31 historical periods, 5000 BCE to 1950 CE
- Compare treated (river shifted toward) vs. control areas
- How much did the shift *increase* state formation?



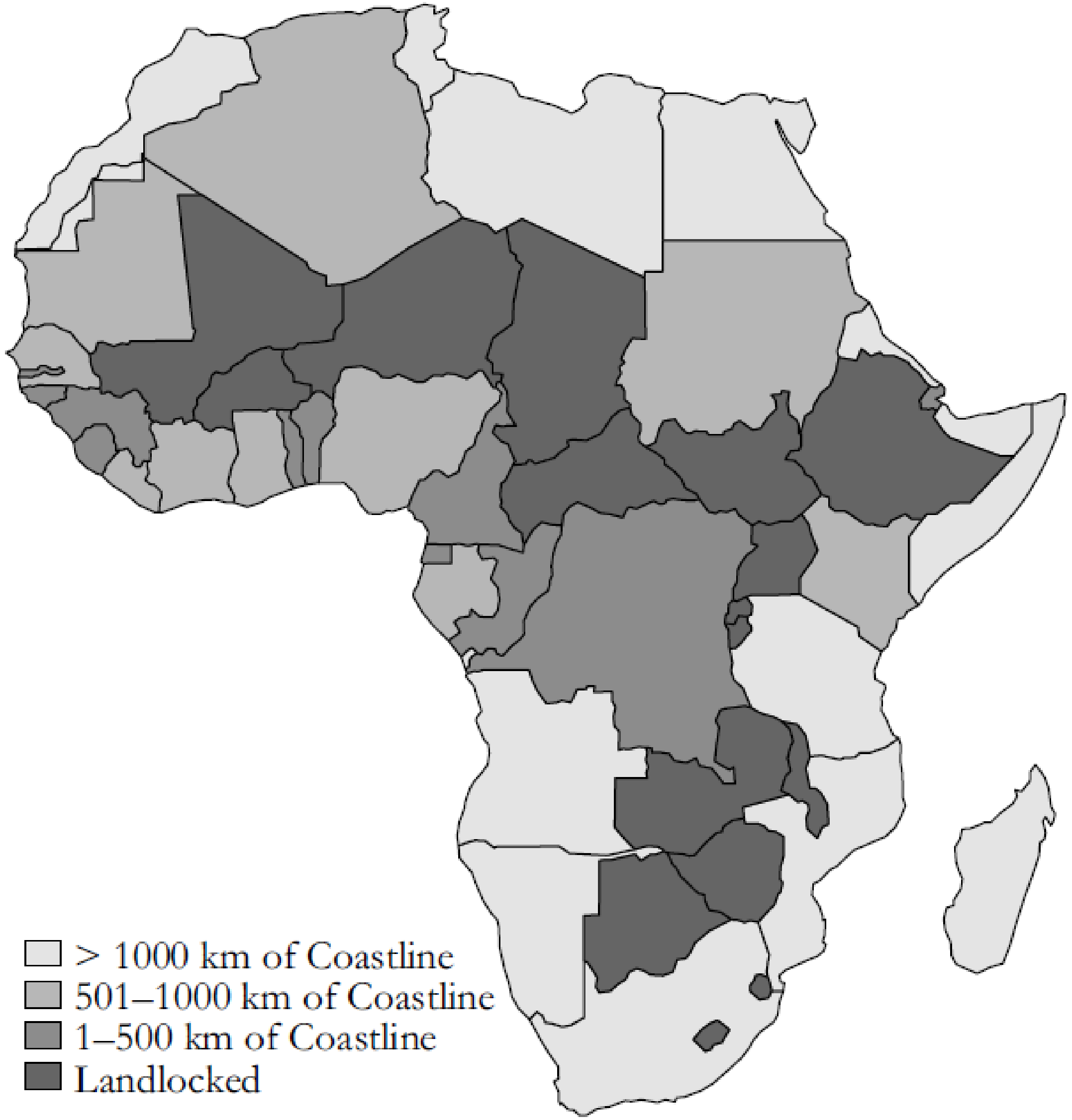

Panel A. Main result
Estimated effect of a river shift on being under a state (yes/no)
0.2
0.15
0.1
0.05
0
River shifts away
Treatment effect
DID-coefficient: 0.14
Comparison mean: 0.24
–3
–2
–1
0
1
2

Machine learning versus econometrics

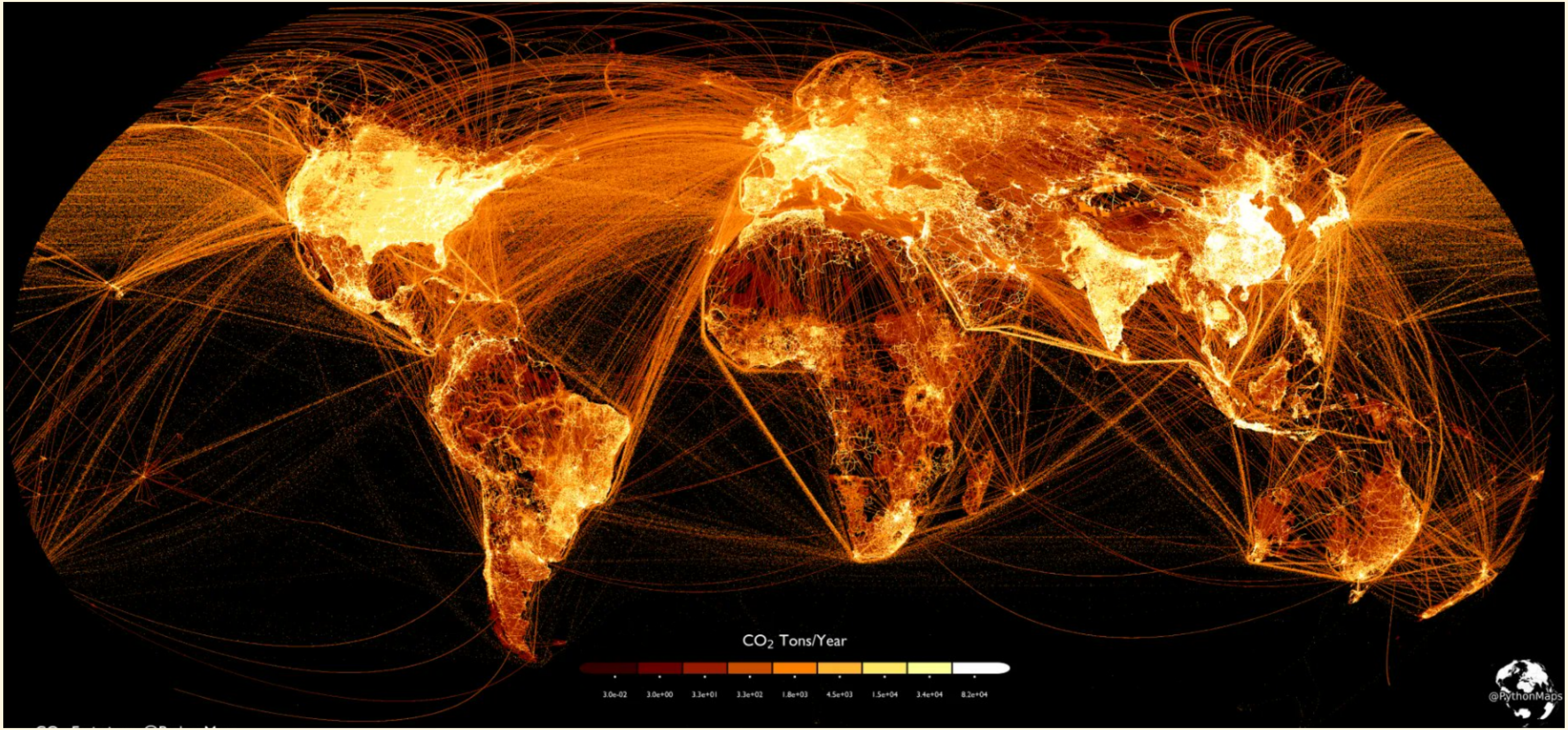
- ML (including neural networks) is a new term for *applied predictive modeling*.
    - ML values robust prediction accuracy.
    - Searches through very flexible non-linear models to find best 'fit'.
    - but generally not for causal identification or unbiased estimate of impacts.
    - If ice-cream helps predict drownings, model will use ice-cream.
    - Not useful for policy: Prohibit ice-cream will not reduce drownings!
- Econometrics is for hypothesis-testing.
    - Economist uses theory and evidence to posit an identified causal relationship
    - Uses statistical inference methods to obtain unbiased estimates of causal model parameters and tests hypothesised relationships.

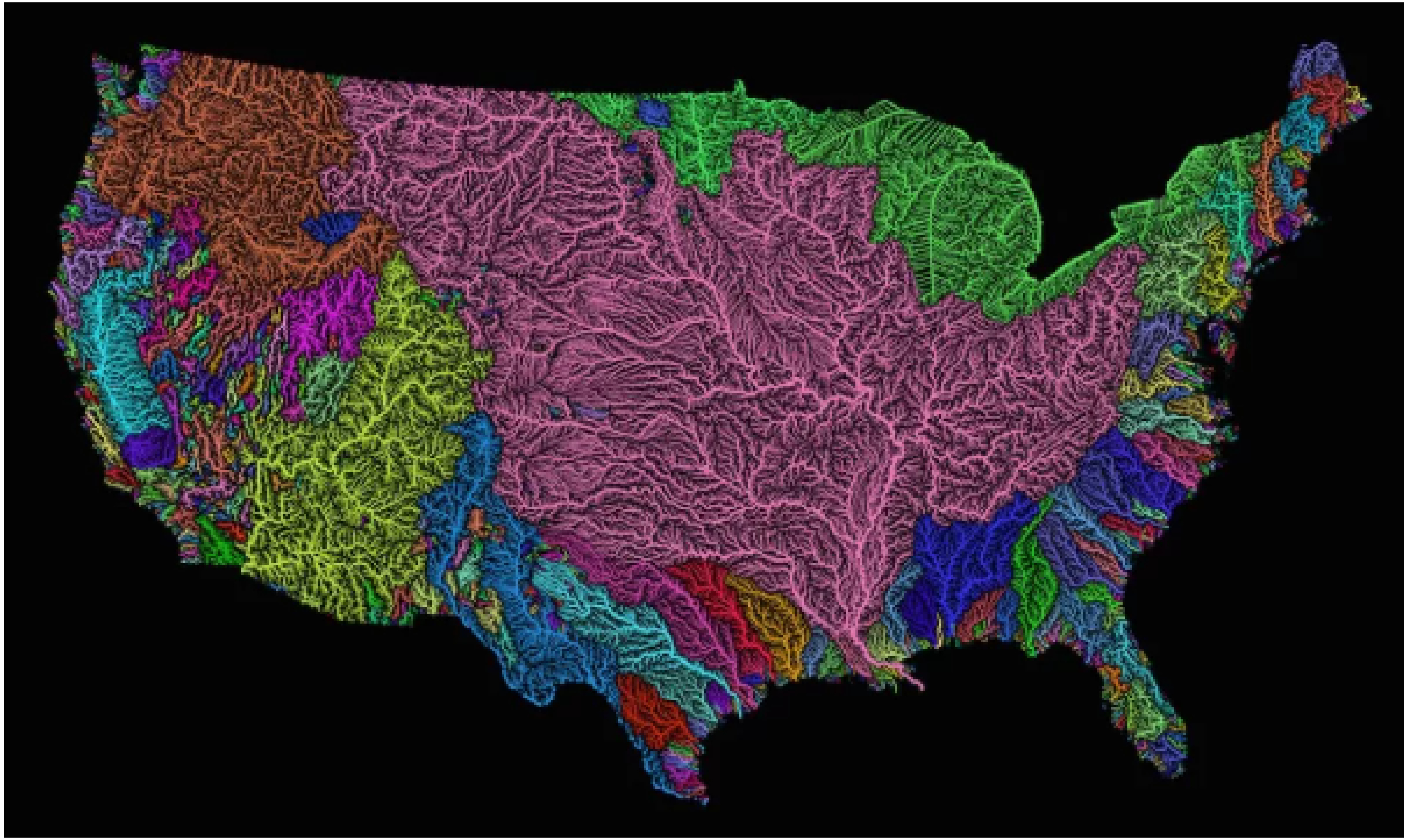
Geography and Economic Development

- How does geography shape development?
- Jeff Sachs (Columbia) observes many poorest countries are landlocked.
- Water-based the most cost-effective transport, so a major disadvantage.



re 2.1 Coastlines of African countries


CO_2 Tons/Year
3.0e-02
3.0e+00
3.3e+01
3.3e+02
1.8e+03
4.5e+03
1.5e+04
3.4e+04
8.2e+04
@PythonMaps



A new map visualizes the flow of ever river in the United States. (Image credit: Robert Szucs, Fejetlenfej/Imgur) next slide:

India's Ganges River Basin

- One of the most fertile places on earth
- Hot (at same latitude as deserts)
- But moist due to monsoon rains off the Arabian sea

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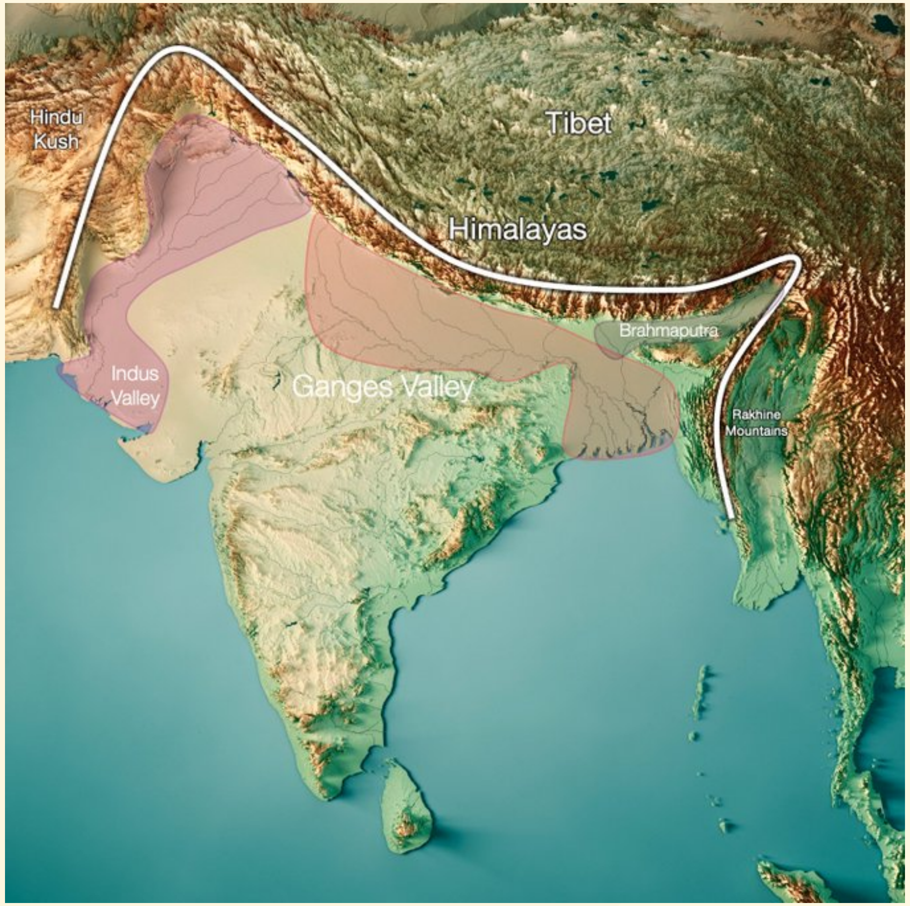
River Basins



[Source]

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[Population through the ages video]

Geography and Disease Environment

- Geography determines disease environment.
- Malaria has likely killed more than any other cause.
- Malaria belt regions grow as much as 1.3% slower (Gallup and Sachs, 2001).
- By reducing population density, delayed state formation.

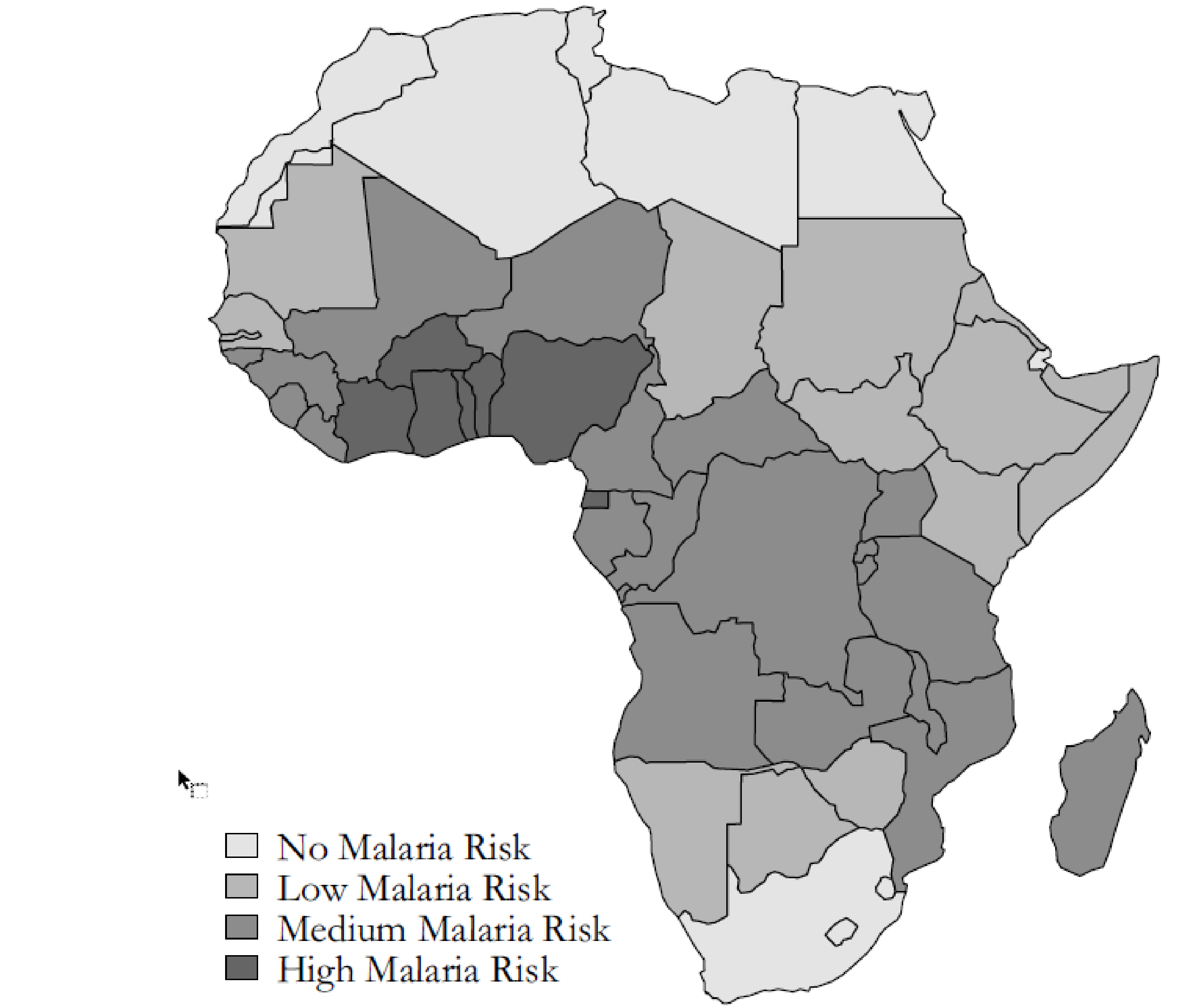


Figure 2.2 Africa's "malaria belt"

Data source: Hay, Guerra, Gething, Patil, Tatem, Noor, et al. (2009).

The Effect of the TseTse Fly on African Development[†]

By MARCELLA ALSAN*

The TseTse fly is unique to Africa and transmits a parasite harmful to humans and lethal to livestock. This paper tests the hypothesis that the TseTse reduced the ability of Africans to generate an agricultural surplus historically. Ethnic groups inhabiting TseTse-suitable areas were less likely to use domesticated animals and the plow, less likely to be politically centralized, and had a lower population density. These correlations are not found in the tropics outside of Africa, where the fly does not exist. The evidence suggests current economic performance is affected by the TseTse through the channel of precolonial political centralization. (*JEL* I12, N57, O13, O17, Q12, Q16, Q18)

Geography and Disease Environment

- tsetse fly causes sleeping sickness.
- kills livestock
- Impacts economic development and political formation
- Where fly is active less likely to develop large-scale states (Alsan, 2014)

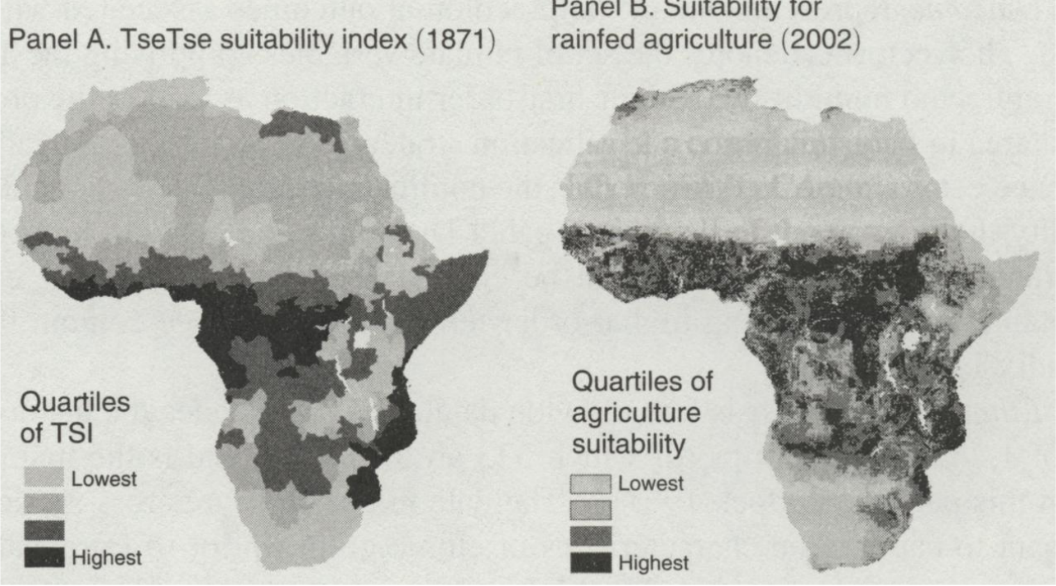
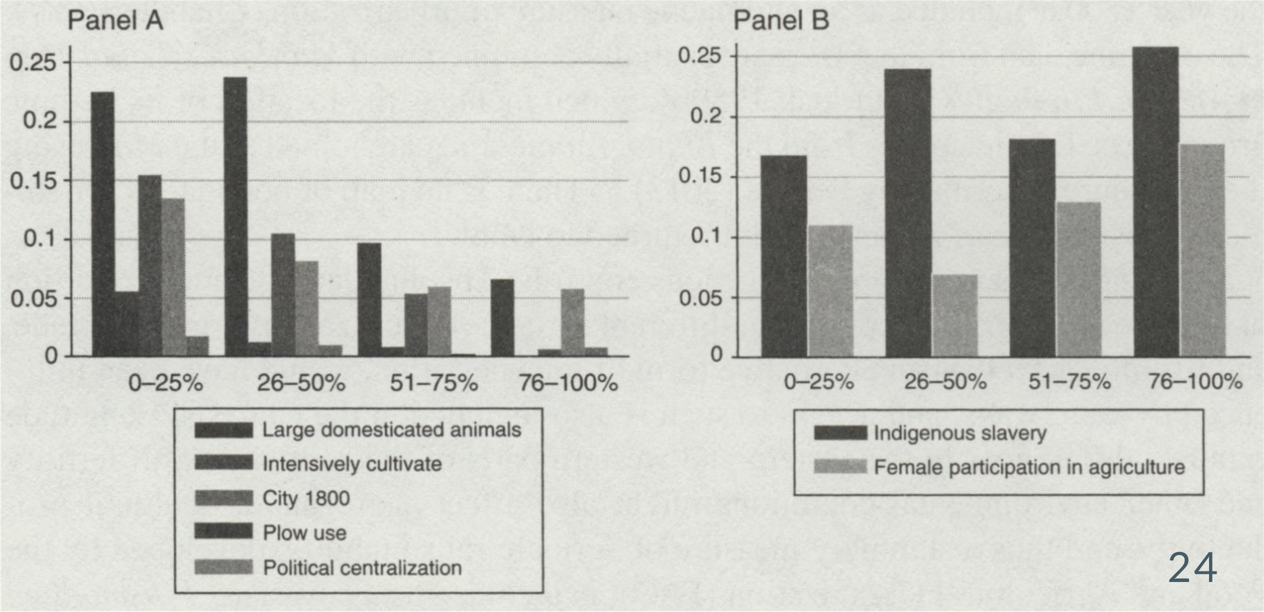


FIGURE 3. TSETSE SUITABILITY INDEX AND THE SUITABILITY FOR RAINFED AGRICULTURE

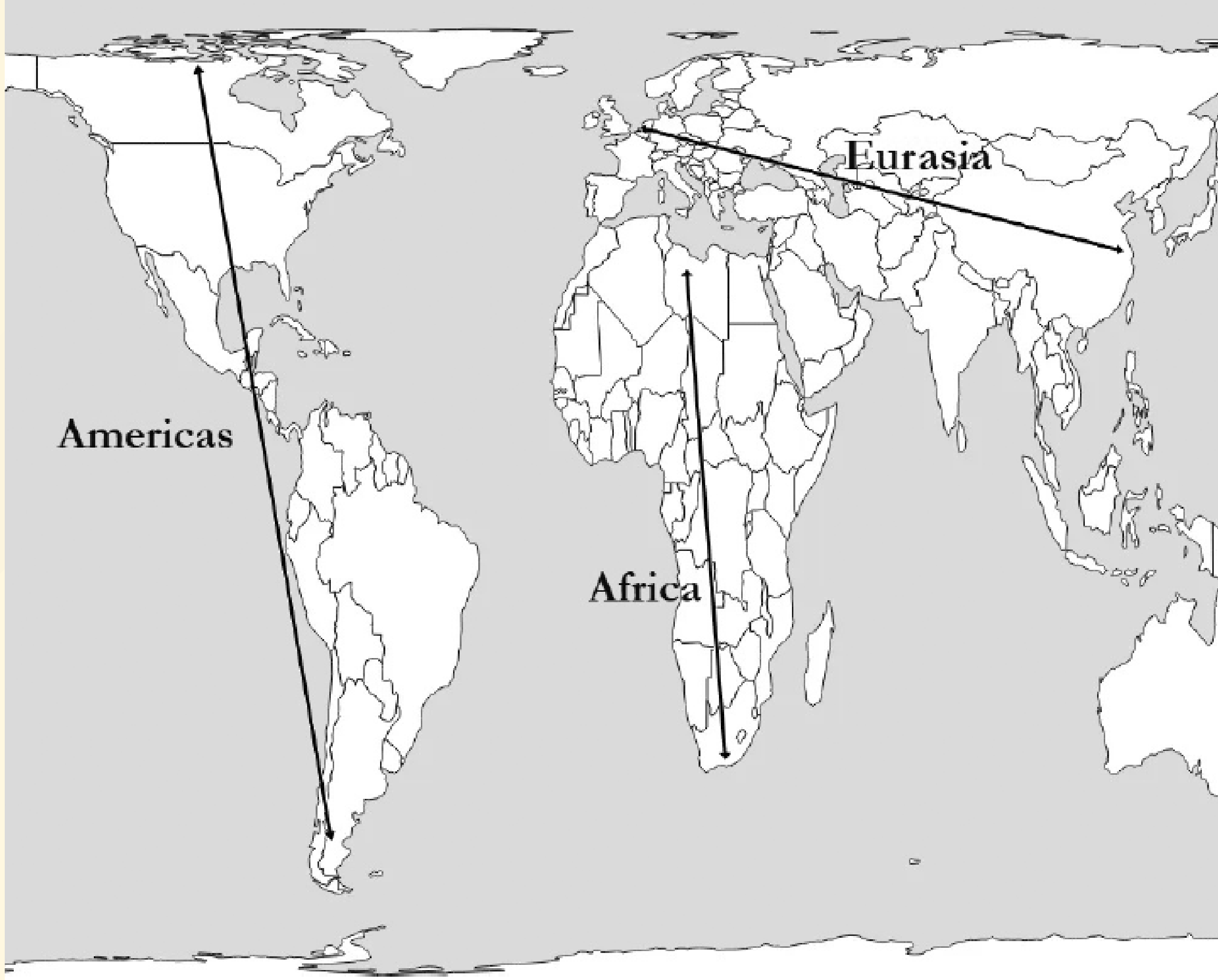
Notes: Panel A shows the historical TseTse suitability index created using climate data from NOAA's 20th Century Reanalysis for the year 1871. Panel B shows the suitability for rainfed agriculture (FAO 2002).



Jared Diamond

Guns, Germs, and Steel

- Shape of Continents determined speed of technological diffusion.



.3 Verticality and horizontality of the continents
iced using map in Diamond (1997). Gall-Peters map projection.

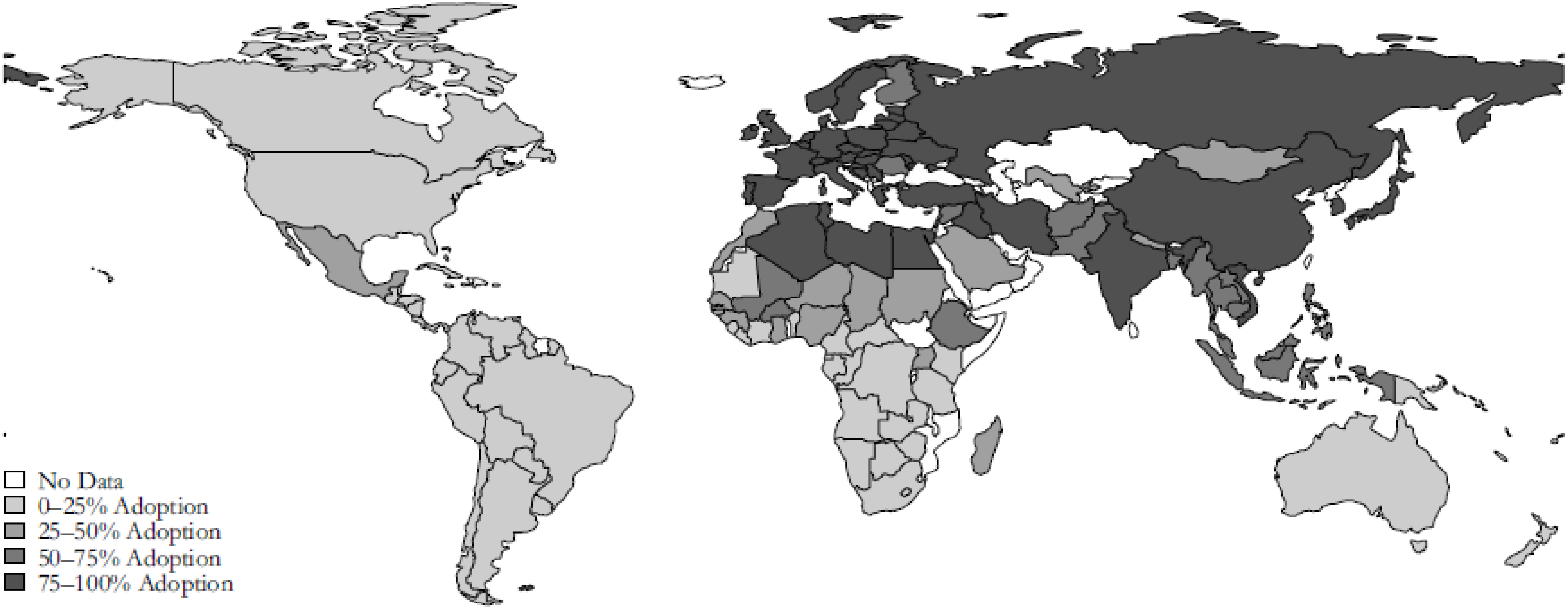


Figure 2.4 Technology adoption levels in 1500 (% of frontier technologies adopted)

Data source: Pavlik and Young (2019).

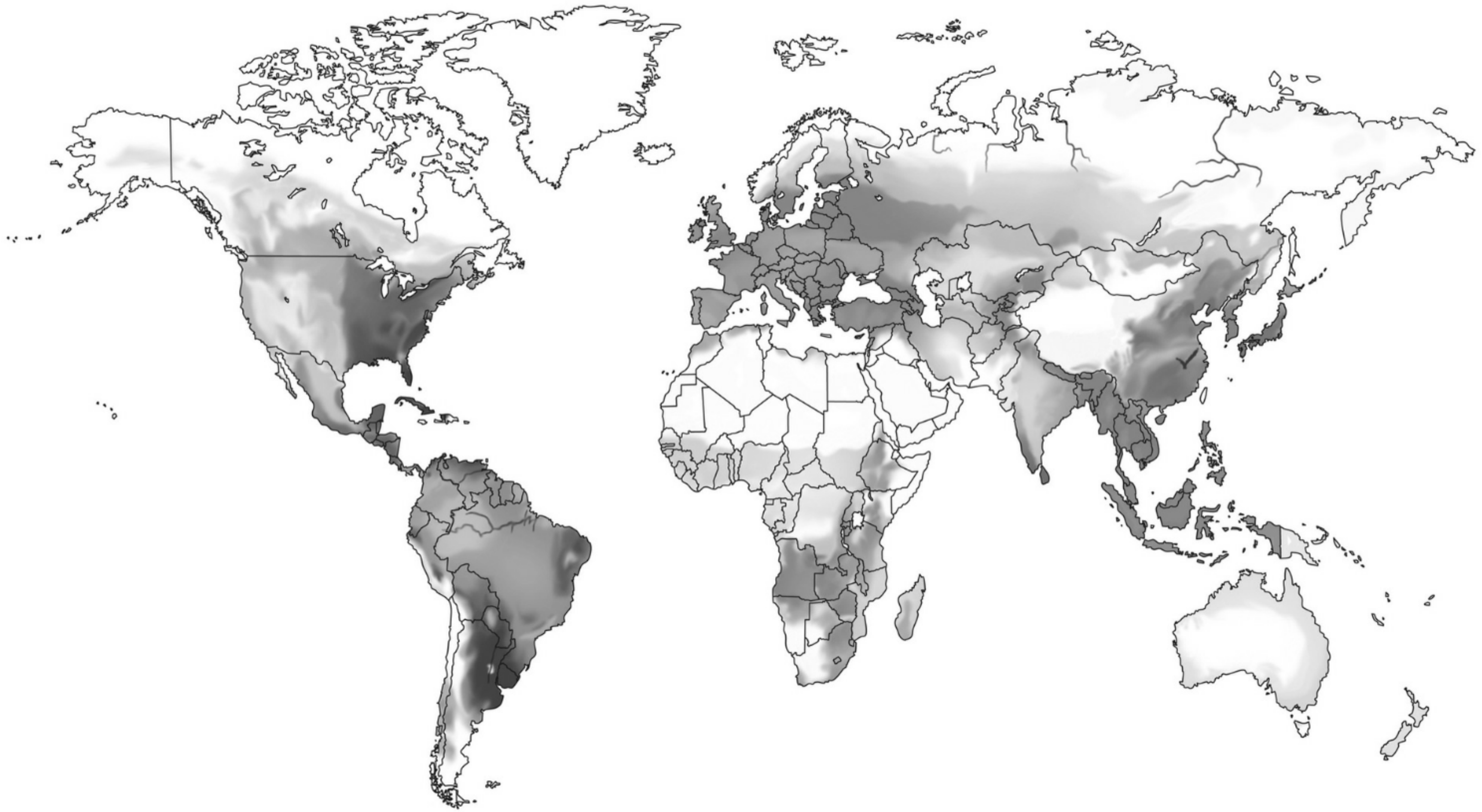
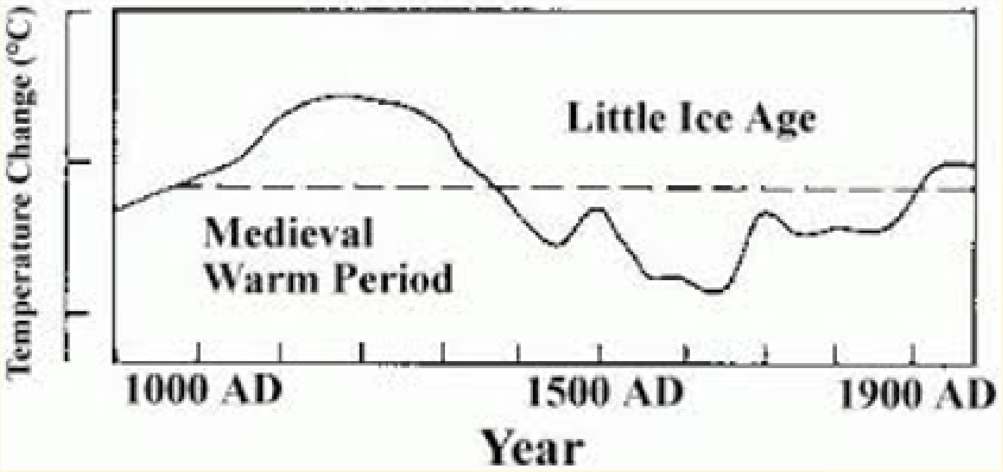


Figure 17. Potential Caloric Return of Native Crops before 1500 CE

This figure depicts the worldwide distribution of daily potential caloric yields, over the period from planting to harvesting, for crops that were native to a location before 1500 CE. Higher values of potential caloric yield are indicated by darker shadings.[28]

Geography and Climate

- Climate changes.
- Roman Empire arose during period of exceptionally warm weather.
  - Decline during colder period
- After 1000 CE warmer weather again, higher yields and growth in W Europe.
- Colder weather spells ⇒ negative econ shocks.
  - Conflict, pogroms...

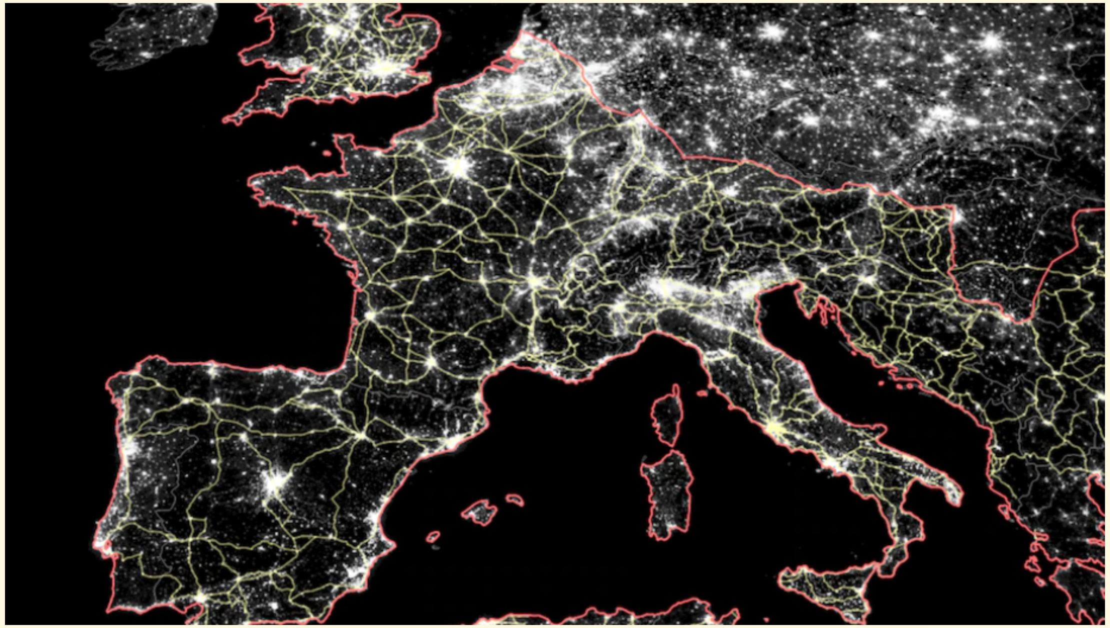


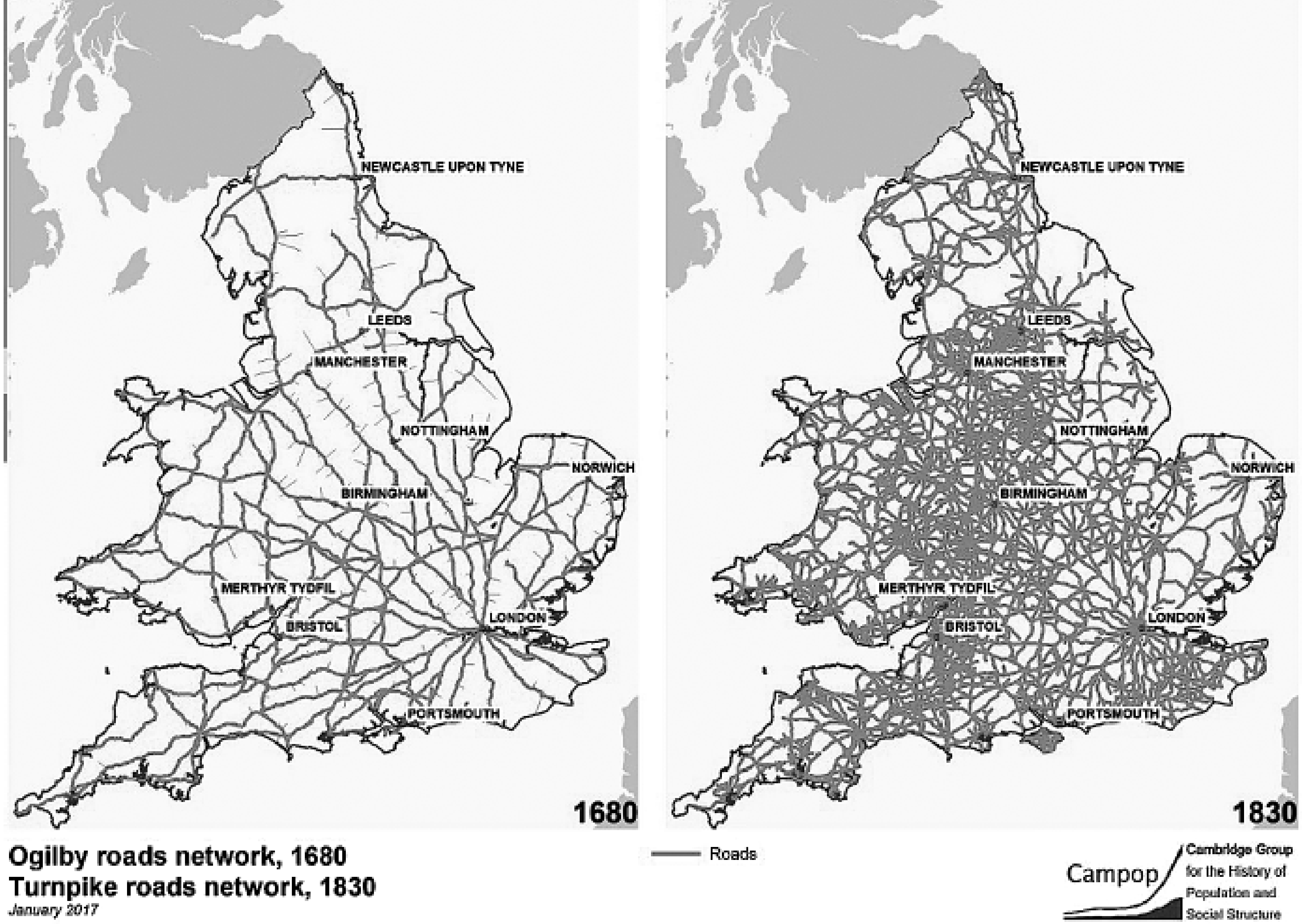
Geography and Transport Infrastucture

- Geography matters but can be adapted
- Investment in transport infrastructure can reduce barriers to trade.
- Roman Road building
- Persistent long effects:
  - European road network until 19th Century
  - Location of cities
  - Positive and Negative effects

Figure 2.6 The Roman road network

Map provided by Erik Hornung from Flückiger, Hornung, Larch, Ludwig, and Mees (forthcoming). Grey lines symbolize roads, solid black lines navigable river sections, and dashed lines coastal shipping routes.





(a) Turnpike Roads in 1680 (b) Turnpike Roads in 1830

Figure 2.7 The increase in turnpikes in England and Wales: 1680–1830

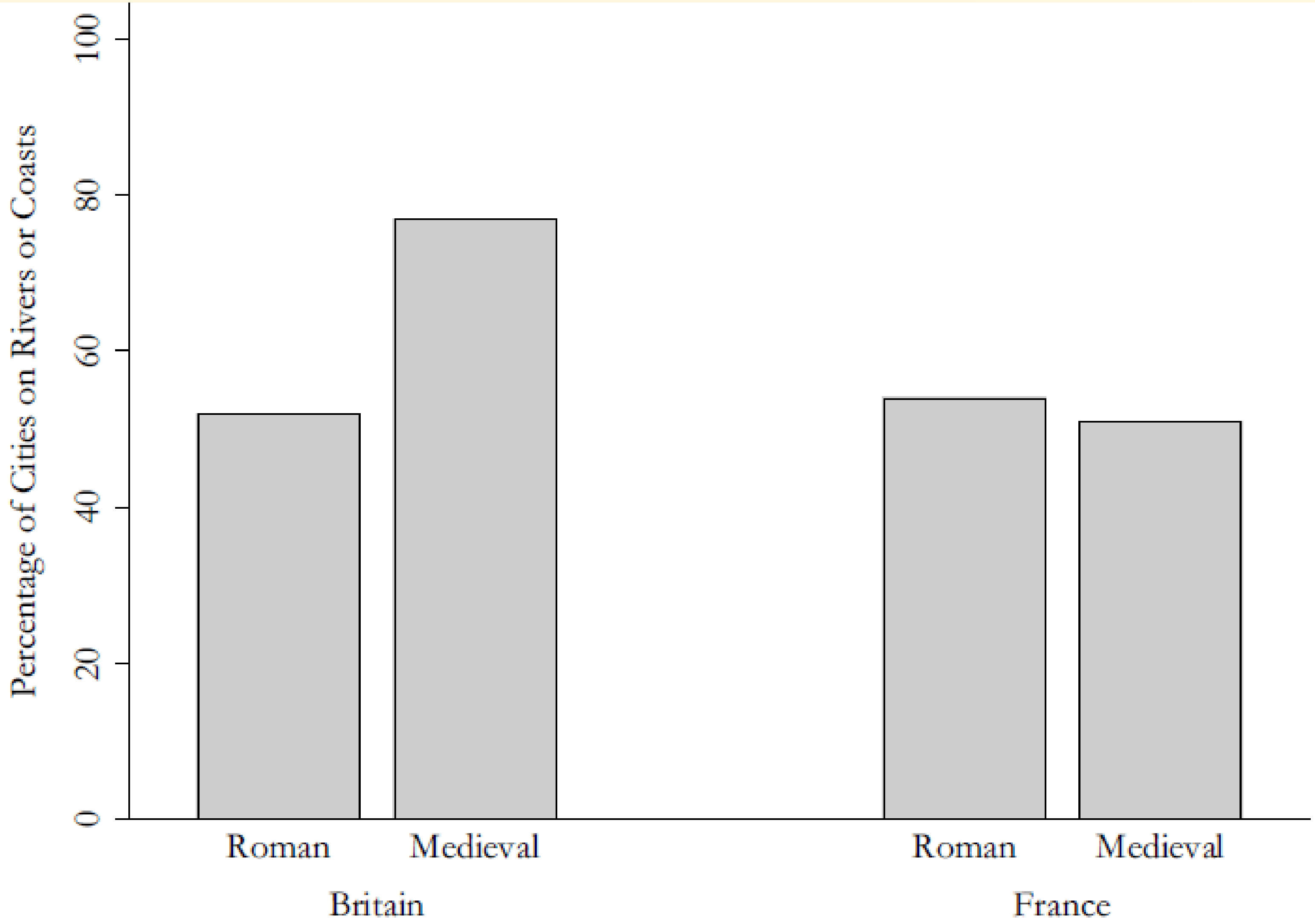


Figure 2.9 Location of cities in England and France in the Roman and medieval periods

Data source: Michaels and Rauch (2018).

The Columbian Exchange

The Potato

(Friday presentation)

Centralization vs Competition

- Landscape fragmentation and the rise of Europe (Hume, Tilly)
    - Competition (vs Incumbent Monopolist)
        - Military
        - Public goods to attract population
        - Entrepreneurs, scientists, shopped regions.
    - Witfogel hydraulic hypothesis. Fractal coastline.
- Is geographical connectivity a benefit or a disadvantage?

Origins of Extractive vs. Inclusive Institutions

- Did agroclimate shape institutions in US South, Latin America?
- Did geography/disease environment affect where colonization left

Geography, Crops, and future-orientation

- Future orientation: People from areas with higher *potential* caloric return appear to be more future-oriented (from surveys).
  - Reverse causality?
- Gender roles
  - Ploughs vs hoes and rakes.
  - World Values Survey suggests gender biases correlate.
    - Even in second-generation immigrants from countries that used the plough.

Next class Readings

- OG 2 "Lost in Stagnation" 27-41
- OG 3 "The Storm Beneath the Surface", 43-55.
- (optional) KR 5 Fewer Babies?

Student focus paper:

- Nunn, Nathan, and Nancy Qian. 2011. “The Potato’s Contribution to Population and Urbanization: Evidence from a Historical Experiment.” *The Quarterly Journal of Economics* 126 (2): 593–650. (PDF)

Population and growth Questions 1

- What is the "Malthusian Trap?"
- What key elements of human demography and economic history does it explain?
- What are key assumptions of the model?
- Steady states and level versus growth effects
- Malthusian comparative statics
    - effect of raising/lowering birth rate?
    - effect of raising/lowering death rate
    - effect of raising/lowering technological productivity
- Date of onset of agriculture and expansion?